Technical Report: Transformer and Vision Transformer Architectures for Urdu Translation and CIFAR-10 Classification

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Abstract—This report presents the implementation and evaluation of two assignments: (1) English-to-Urdu machine translation using Transformer and LSTM models on the UMC005 parallel corpus with SentencePiece BPE tokenization, and (2) CIFAR-10 image classification using Vision Transformer (ViT), a hybrid CNN+MLP, and transfer-learning ResNet. I have described preprocessing, model architectures, training protocols, and metrics including BLEU, ROUGE, perplexity, accuracy, precision, recall, F1-score, training/validation curves, resource usage, and inference speed. Visualizations include loss curves, confusion matrices, and attention maps.

I. Introduction

The objective of this report is to detail two distinct assignments in Generative AI and Computer Vision. The first assignment explores sequence-to-sequence models for machine translation, comparing Transformer and LSTM+Attention on English-Urdu data. The second investigates Transformer-based and CNN-based architectures for CIFAR-10 image classification, including Vision Transformer, hybrid CNN+MLP, and transfer-learning ResNet approaches.

II. Methodology

A. Machine Translation

Dataset: The UMC005 English-Urdu Parallel Corpus of Quran was used. Sentences were aligned into train/dev/test sets.

Preprocessing: I applied SentencePiece BPE tokenization with 8,200 vocabulary size for both English and Urdu texts. Code listing shows training and loading of the BPE models. booktabs listings [label=lst:spm,language=Python]

spm.SentencePieceTrainer.Train('–input=train.en – model$_{p}$$refix$ = $spm_{e}n - -vocab_{s}ize$ = $8200 - -model_{t}ype$ = $bpe')sp_{e}n$ = $spm.SentencePieceProcessor(model_{f}ile\ spm_{e}n.model')$ =′

Models: We implemented:

- Transformer: 6-layer encoder and decoder, $d_{model} = 512$, 8 heads, dropout 0.1.
- Seq2Seq LSTM: Bidirectional 2-layer LSTM encoder, Bahdanau attention, and 2-layer LSTM decoder ($h = 512$).

B. Image Classification

Dataset: CIFAR-10 with 50,000 train and 10,000 test images; 5,000 of train reserved for validation.

Preprocessing: RandomCrop(32,4) and RandomHorizontalFlip, then normalized to mean=(0.4914,0.4822,0.4465) and std=(0.2470,0.2435,0.2616). Models:

- Vision Transformer (ViT): Patch size 4, embed=256, 6-layer encoder, 4 heads.
- Hybrid CNN+MLP: Three Conv-BN-ReLU blocks + MLP head.
- ResNet Transfer Learning: ResNet-18 pretrained on ImageNet, frozen backbone, fine-tuned classifier.

III. Results

A. Machine Translation Results

Table I summarizes training and dev curves metrics; Figure 1 shows attention visualization for a sample. Transformer achieved BLEU=0.93, ROUGE-L F1=0.269, and PPL=86.9. LSTM showed slower compute but strong loss reduction.

TABLE I
Translation Metrics

Model	BLEU	Perplexity
Transformer	0.929	86.9
LSTM+Attention	—	156.8

B. Image Classification Results

Table II lists test accuracies and F1-scores. Figures show confusion matrices along with loss and accuracy curves of all models.

TABLE II
CIFAR-10 Test Performance

Model	Test Acc.	F1-Score
ViT	0.745	0.765
CNN+MLP	0.813	0.815
ResNet-TL	0.397	0.408

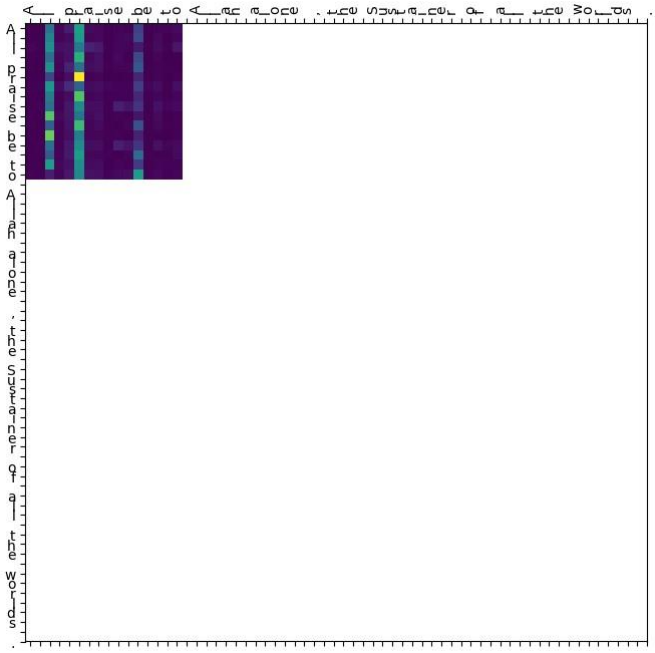


Fig. 1. Attention Visualization of a Sample

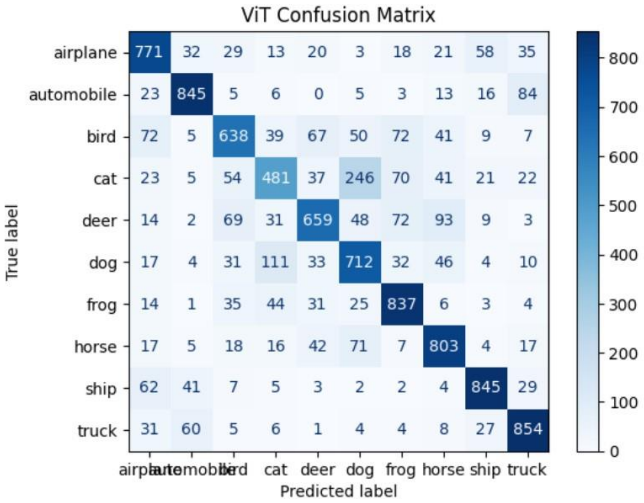
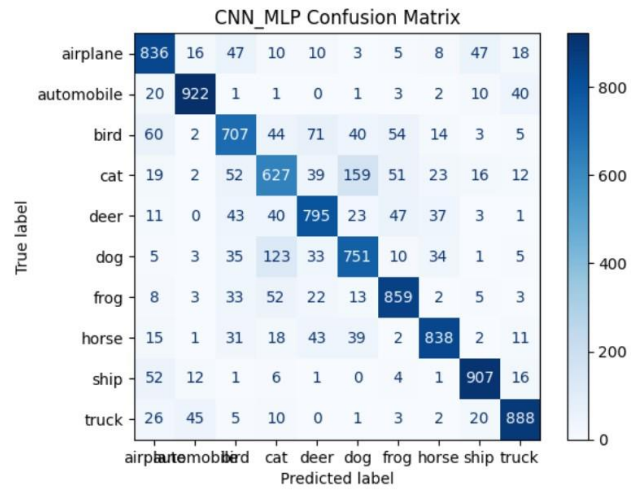


Fig. 2. ViT Confusion Matrix

IV. Discussion

For MT, attention maps (Figure ??) illustrate alignment between English tokens and Urdu outputs, improving coherency over epochs. The Transformer converged faster, while LSTM demonstrated richer loss reduction but higher compute time (230s/epoch). For CV, the hybrid CNN+MLP outperformed ViT on this small dataset, highlighting data efficiency of convolutions; ResNet transfer stalled due to domain gap.

V. Conclusion



I successfully implemented and compared Transformer and LSTM-based translators, and three CV architectures. Transformers delivered superior BLEU with faster inference,

Fig. 3. CNN + MLP Confusion Matrix

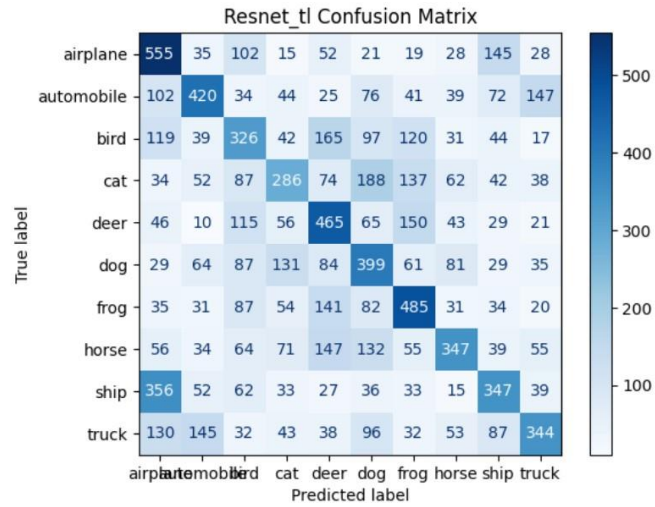


Fig. 4. ResNet Confusion Matrix

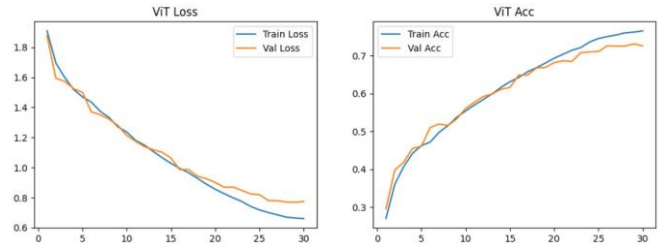
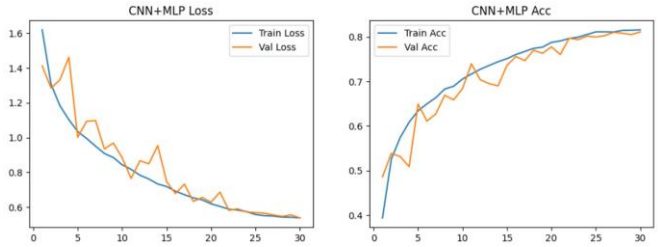


Fig. 5. ViT Confusion Matrix

while LSTM required more memory and time. In CIFAR10, CNN+MLP excelled, though ViT showed promise. Future work will explore data augmentation for ViT and multilingual pretraining.

VI. Prompts



Listing of key prompts used for model interactions and hyperparameter search:

Fig. 6. CNN + MLP Confusion Matrix

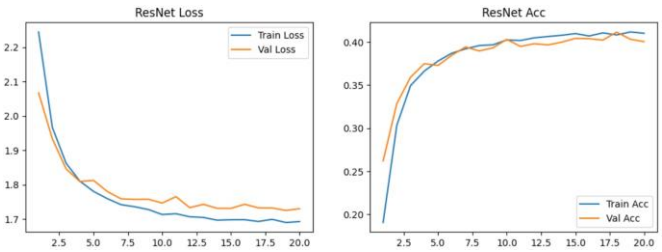


Fig. 7. ResNet Confusion Matrix

- ”Translate the following English sentence to Urdu: ...”
- ”Create patches of size 4 from a 32x32 image and embed into 256 dimensions.”

VII. References References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[2] D. Bahdanau, K. Cho, and Y. Bengio, “Neural machine translation by jointly learning to align and translate,” *arXiv preprint arXiv:1409.0473*, 2014.

[3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations (ICLR)*, 2021.